Sepehr Vahedi

Software Engineer (3.5 years) | ML Infrastructure, Model Serving, Backend

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Right to work: available from Sep 2027 on a UK Graduate visa (18 months, no sponsorship needed to start)

EDUCATION

MSc Software Systems Engineering — Sep 2026 – Sep 2027 (expected)

University College London (UCL) — *London, UK*

- Modules: Software Architecture, Security and Privacy of AI, Financial Institutions and Markets, Industrial Group Project.

BSc Computer Engineering (Software Engineering) — Sep 2020 – Jul 2026

Sharif University of Technology — *Tehran, Iran*

- GPA 16.71/20 (**18.25/20 in the final two years**); six years due to co-op and full-time work at Tapsi, then 2025 in research.

EXPERIENCE

Tapsi – *ride-hailing and delivery super app, 10M+ monthly active users* — *Tehran, Iran*

Senior Software Engineer, Delivery and Marketplace — Dec 2025 – Sep 2026

- Raised ETA model-server throughput **54x, from 70 to 3,800 requests/sec per pod**, and cut p50 latency at peak from 866 ms to 62 ms by removing a Triton serialization bottleneck and batching sub-model calls; outputs verified identical on 19,200 requests.
- Took the delivery domain's first ML models to production on Triton (KServe v2 gRPC, CI/CD to production, accuracy gates); designed the six-city shadow and switchback A/B rollout, balanced for time of day: **ETA MAPE fell from 24% to 19%**.
- Designed and owned the delivery-routing gRPC service (Kotlin, Spring WebFlux) behind every motorcycle ETA and distance estimate (ML first, then OSRM, then Haversine); migrated five services off the legacy Go estimator with no regressions.

Senior Software Engineer — Jan 2024 – Dec 2024

- Built the event-sourced Kafka backend (1M+ events/day) for multi-delivery orders; the launch **doubled daily deliveries**.
- Rewrote the ride-request service, which manages the lifecycle of every ride request, from a Node.js monolith to Spring Boot and Project Reactor with zero downtime, cutting its infrastructure cost **45%** and peak response time from 200 ms to 80 ms.

Software Engineer — Feb 2022 – Dec 2023

- Cut GPS-service load **67%** with adaptive driver GPS intervals, caching and filtering, and rewrote reverse geocoding in Go.
- Built the backend for a new business line, Iran's first school-transport platform (**2,000+** drivers, **12,000+** students), as one of two backend engineers on a 1.5-month deadline; designed its multi-app architecture and offline route matching.

RabinCash | Co-Founder and Product Designer — Feb 2020 – Feb 2021

- Co-founded a P2P and B2P crypto exchange (**20+ trading pairs**); owned its product and UX design from concept to launch.
- Led a two-person front-end team that built the web trading platform (JavaScript, HTML/CSS, Bootstrap).

RESEARCH EXPERIENCE

TestCouncil: Multi-Agent LLM Unit-Test Generation (BSc thesis) — Aug 2025 – Jul 2026

Software Quality Research Lab, Sharif University; supervisor: Prof. Seyed-Hassan Mirian-Hosseinabadi — *Tehran, Iran*

- Qwen, DeepSeek and Gemini agents plus an LLM synthesis step that removes ∼85% of redundant tests: **84.6% line and 92.5% branch coverage** on 135 HumanEval, MBPP and algorithm functions, **36+ points** above the best single model ($p < 0.001$).

LLM Issue-Title Generation — Mar 2025 – Feb 2026

Intelligent Software Engineering Lab, Sharif University; supervisor: Prof. Abbas Heydarnoori — *Tehran, Iran*

- Cut LLM input-token cost **37.3%** via learned context selection (2.6M+ GitHub-issue dataset), within 0.004 BERTScore of GPT-4.

PROJECTS

- **Mesh-level rate limiter** (self-initiated at Tapsi): per-user, per-API token-bucket limits as an Envoy Lua filter on Istio, adopted by every microservice exposed to external clients; part of my promotion case.
- **Property platform** (Air Properties, Dubai): ETL for 10K+ listings; RAG recommender, 500+ queries/day, **40% faster matching**.
- **Multi-agent paper critique**: four LLMs cross-rank each other's critiques (shuffled against position bias), merged by Borda count.

LEADERSHIP AND MENTORING

- **Tapsi:** led the delivery-routing and ML-serving projects end to end with a data scientist, owned code review and on-call for the routing stack, and was the sole onboarding mentor through a new engineer's first three months.
- **Teaching Assistant**, Sharif: Advanced Programming (2024); Scientific and Technical Presentation (2025–26).
- **Informatics Olympiad Teacher**, NODET (2020–21): my students won **one gold and two silver INOI medals**.

SKILLS

Languages	Kotlin, Java, Python, Go, C++, TypeScript and JavaScript (Node.js), SQL
ML Serving	NVIDIA Triton Inference Server, KServe v2 protocol, ONNX, PyTorch, Hugging Face Transformers, LLM APIs
Backend	Spring Boot, Project Reactor (WebFlux), gRPC, REST, Apache Kafka, event-driven microservices, JUnit, Mockito
Infrastructure	Kubernetes, Docker, Helm, ArgoCD, GitLab CI/CD, Istio, Envoy, Prometheus, Grafana, Linux, Git
Data	PostgreSQL, MongoDB, Redis, Hazelcast, S3-compatible object storage, Uber H3, OSRM
Spoken	English (IELTS 8.0), German (intermediate), Persian (native)

AWARDS

- **Codeforces Candidate Master** (max 2075, top 1.5%); Expert (max 1765) on my earlier account; 420+ problems solved.
- **Iran National Olympiad in Informatics**, top-50 finalist (2018, 2019); **AUT International Programming Contest**, 3rd of 65.